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MAGNETIC PROPERTIES OF THE CUNENE ANORTHOSITE COMPLEX, ANGOLA

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The Cunene Complex is the largest known anorthosite body and outcrops across the border between Angola and South West Africa. Palaeomagnetic results are reported from a traverse across the dark troctolitic facies of the anorthosite in Angola which yielded fifteen sites with two additional sites in gabbro bodies. Fourteen sites are stable to a.f. demagnetisation and a single site in the cumulative border zone of the anorthosite is reversed with respect to the remainder. Twelve sites combine to give a mean direction of $D = 259°$, $I = -46°$ ($k = 7$) with a virtual geomagnetic pole at 255°E and 3°S. The low overall precision is probably due to apparent polar movement during cooling of the Complex. Radiometric data are currently conflicting and imply that the anorthosite has an age between 1100 and 2600 m.y.; the only clear feature to emerge from age studies is a thermal overprinting at ca. 1100 m.y. The directions of magnetisation are shown to be most consistent with an age of ca. 2100 m.y. with cooling through the Curie point continuing to ca. 2000 m.y.

A variety of magnetic tests demonstrate that magnetite is the principal remanence carrier in the dark troctolitic anorthosite where it occurs both as discrete grains and as fine rods in plagioclase. Lowrie-Fuller tests suggest that both these components include single domains but results from separated mineral fractions demonstrate that the bulk of the high coercivity remanence resides in magnetite rods within the feldspar.

1. Introduction

The Cunene Complex is the largest known body of anorthosite on the globe with an exposed outcrop of 12,500 km^2, and a total area of perhaps 16,000 km^2 if probable extensions of the outcrop below a thin sedimentary cover are taken into account (Simpson, 1970). The Complex forms an arcuate outcrop (Fig. 1) straddling the border between South West Arica and Angola, and is cut across by the Cunene River near its southern margin.

The ealiest important description of the Complex was given by Mouta and O'Donnell (1933) and maps of Mouta (1954, 1955) help to clarify field relationships of the anorthosite to surrounding rocks. Montenegro de Andrade (1950) has also described the Complex and Stone and Brown (1958) give petrographic details. Geological knowledge to the end of the last decade has been summarised by Simpson (1970). In Angola the Complex is currently being mapped by the Instituto de Investigaçao Cientifica de Angola, and this paper reports magnetic properties of a palaeomagnetic collection sampled across the Complex here. The collection forms a sample through most of the width of outcrop (Fig. 1) and was made on or close to the new road linking Luanda and Windhoek.

2. Geologic outline

The anorthosite intrudes a variety of country rocks metamorphosed to albite–epidote–amphibolite or amphibolite facies (Simpson, 1970) including granites, granite gneisses, amphibolites, metasediments (after limestones and quartzites) and metavolcanics. The contact appears to be discrete in the north but in the south the anorthosite is intimately associated with patches of country rock, and contact metamorphic minerals including cordierite and garnets are widely developed in the latter.

A range of intrusions including granites, granodiorites, syenites and pegmatites intrude the Complex over most of its area. It is overlain unconformably by near-horizontal sediments of the Bembe System forming a large outcrop which extends to the northwest. Possible lateral equivalents of the Bembe System in South West Africa

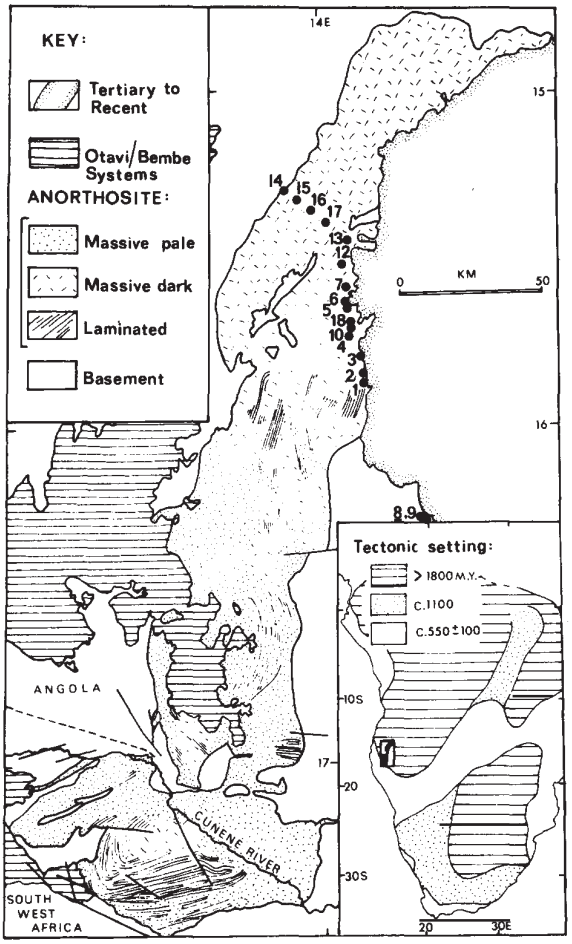


Fig. 1. Simplified geological map of the Cunene anorthosite Complex in Angola and South West Africa based on Simpson (1970) with the positions of palaeomagnetic sampling sites. The inset diagram shows the relationship of the Cunene complex to the cratonic pattern of southern Africa after Clifford (1968).

are correlated with the Nosib Formation and Otavi Group of the Damara System (Martin, 1965; Simpson, 1970). This latter System is assigned to the Late Precambrian; present data suggest a maximum age of 850 m.y. (Kröner, 1971) with sedimentation continuing possibly to ca. 600 m.y. Both the anorthosite and the Bembe System are intruded by dolerite sills and the Complex is extensively intruded by dolerite dykes with trends varying from NE to NW in various parts of the outcrop. These are thought to be feeders of the sills (Simpson, 1970) and probably have a Late Precambrian or Karroo age. The eastern part of the anorthosite outcrop is covered by a thin veneer of Tertiary to Recent conglomerates and sandstones belonging to the Kalahari Beds.

Simpson (1970) recognises three principal facies in the anorthosite body; a dark troctolitic anorthosite which is mostly massive but locally laminated, a pale massive anorthosite, and a mafic or ultramafic marginal facies interpreted as accumulates formed near the floor zone of the anorthosite. The pale massive anorthosite forms much of the southern outcrop of the Cunene Complex but is absent in the north; it is composed of plagioclase of intermediate composition (An_{50-66}) with occasional pyroxene and sometimes olivine. The entire northern part and much of the southern part of the Complex are occupied by dark grey to black troctolitic anorthosites. Lamination is well developed particularly near line of latitude 16° but dips vary from high angles to nearly horizontal; in the outcrop in South West Africa the laminations define a fold-like structure within the anorthosite body (Simpson, 1970) but elsewhere no simple pattern emerges. The troctolitic anorthosite includes plagioclase ranging from An_{54} to An_{80} with amounts usually of the order of a few percent of olivine, orthopyroxene, clinopyroxene and accessory magnetite and biotite; according to Simpson, the mafic content is typically 10–15% with olivine predominant. The relationship between the pale massive and dark troctolitic anorthosites is uncertain but there do appear to be transitional varieties and in one region the latter is clearly intrusive into the former.

Magnetite is a ubiquitous accessory mineral in the troctolitic anorthosite and approaches the status of a major constituent in some regions where it may compose 20% of the rock. Preliminary analyses quoted by Simpson of magnetite concentrates from these rocks indicate a high titanium content (13.4–19.7% compared with 49.4–52.3% $FeO + Fe_2O_3$).

Fourteen of the seventeen palaeomagnetic sites are in the dark troctolitic facies of the anorthosite Complex, one (No. 14) is from the ultramafic cumulative facies and two (8 and 9) are in gabbro bodies near Cahama outside of the main outcrop of the Complex.

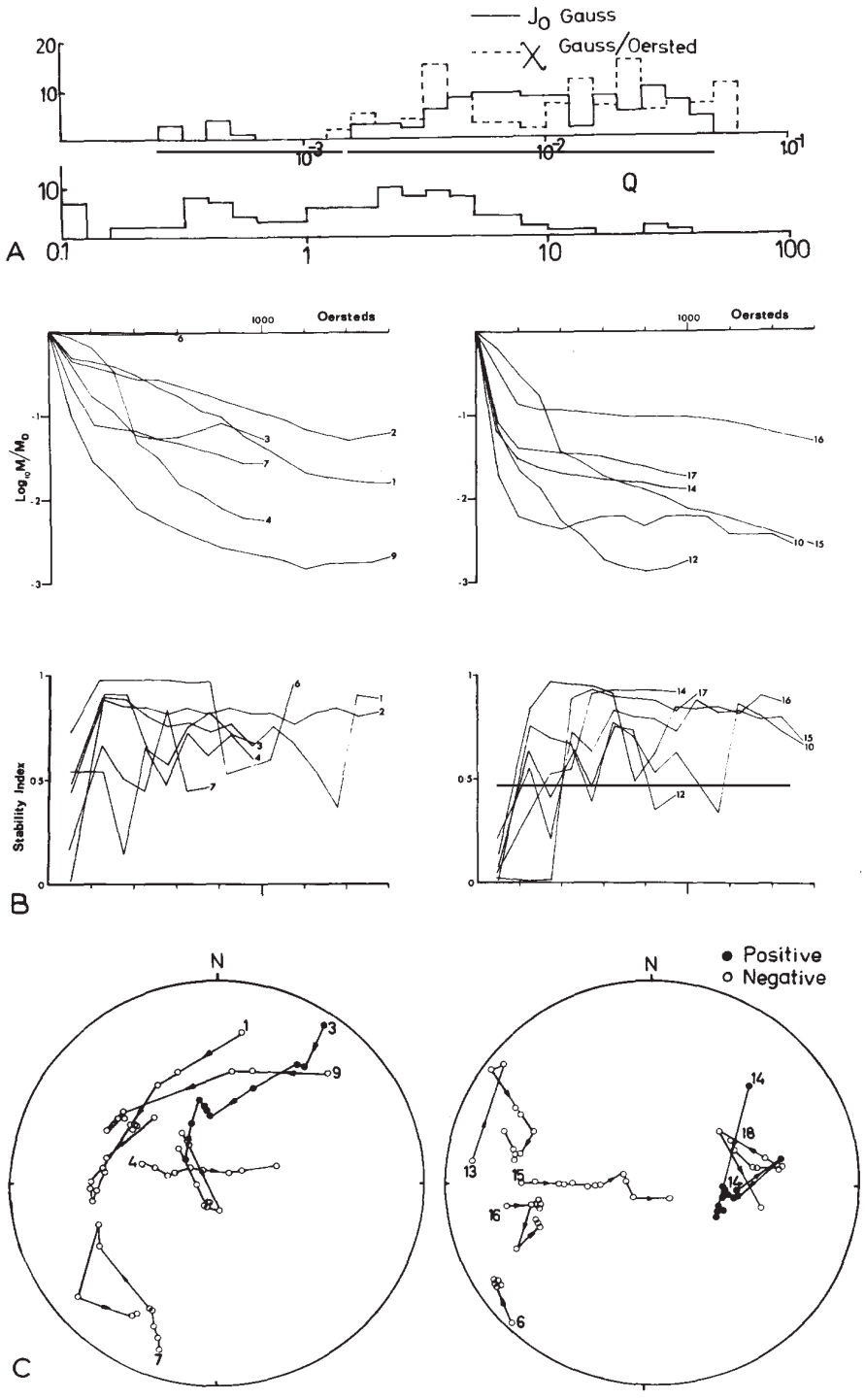


Fig. 2. A. Distribution of intensity of magnetisation (J_0), susceptibility (χ) and Koenigsberger ratio (Q) for samples of the dark troctolitic anorthosite. B. Demagnetisation curves for pilot samples and behaviour of stability indices (Briden, 1972). C. Directional behaviour of some pilot samples.

TABLE I

Site mean directions of magnetisation, Cunene anorthosite Complex, Angola

Site no.	Total NRMs						Cleaning field (s) (Oe)	After a.f. cleaning						Palaeomagnetic pole position	
	N	D	I	R	k	α_{95}		N	D	I	R	k	α_{95}	(°E	°N)
1	9	350	−60	8.1	9	18	400	8	335	−78	7.6	19	13	206	36
2	6	118	−76	5.0	16	17	400–600	6	99	−87	5.6	13	19	188	14
3	7	28	8	6.3	9	21	700–1400	7	1	40	5.5	4	35	15	52
4	4	287	−55	3.4	5	44	1000–1400	5	83	−20	4.0	4	43	112	10
6	6	243	−20	5.4	8	26	500	6	243	−16	5.6	12	20	267	−24
7	4	195	12	3.2	4	57	500	5	223	−30	4.9	44	12	249	−38
8	6	323	−86	5.1	6	30	700	6	214	−60	5.2	6	30	222	−25
9	8	24	−27	4.6	2	52	800	6	277	−31	4.6	4	40	269	11
10	6	not significant					700–1300	6	226	−61	5.4	9	24	229	−18
11	6	not significant								not significant					
12	6	not significant								not significant					
13	7	292	0	4.5	4	36	800	7	264	− 7	4.4	2	52	279	− 5
14	8	28	−37	7.3	10	18	800	8	95	43	6.8	6	25	81	−11
15	7	248	−31	4.8	3	46	600	7	280	−36	6.0	6	26	267	14
16	5	256	− 8	4.7	16	20	400	5	260	−29	4.9	37	13	267	− 5
17	6	not significant					40	not significant							
18	7	not significant					400–1000	8	111	−20	6.5	5	29	121	−17
Overall mean of a.f. cleaned sites excluding sites 3 and 18 and reversing site 14:								12	259	−46	10.5	7	17	255	− 3

N = number of samples; R = length of resultant of N unit vectors; D= declination and I = inclination of remanent magnetisation; $k = (N - 1)/(N - R)$ = estimate of Fisher's κ parameter; α_{95} = semi angle of cone of 95% confidence for mean direction.

3. Palaeomagnetic results

Five lump samples were collected from each site and oriented by both sun sights and magnetic bearings. The sun sights were used to derive orientation data and magnetic bearings were merely used as a check. One to three one-inch diameter cores were drilled from each of the lumps in the laboratory and their magnetic remanence directions were measured before and after treatment in alternating fields using a PAR SM-2D magnetometer. Total Natural Remanent Magnetisations (NRMs) of the samples range from $6.1 \cdot 10^{-2}$ to $3.0 \cdot 10^{-4}$ gauss; they approximate to a log normal distribution (Fig. 2A). Two samples from each site were demagnetised in steps of 100 oersteds (Oe) to 1000 or 1600 Oe and a stability index was computed for each step using the method of Briden (1972).

The behaviour of magnetic moments to a.f. demagnetisation is uniformly similar throughout the collection: samples exhibit sharp falls in moment during the first few hundred oersteds of treatment followed by little or no fall in moment thereafter (Fig. 2B). This is reflected in behaviour of stability indices which show initial large increases after which they oscillate about values between 0.5 and 1.0. Directional behaviour is more complicated (Fig. 2C). No samples showed stable behaviour in fields below about 400 Oe, although pilot samples from the bulk of the sites move to a stable end point either rapidly or more slowly correlating crudely with the rate of fall in moments. For these samples it was possible to define a cleaning field applicable to most or all of the remaining samples of the sites. Samples for sites 3, 4, 6, 10 and 13 did not move to a single stable end point with a.f. demagnetisation and all samples from these sites were accordingly demagnetised to determine the best grouping of directions; in general however these sites finished with lower within-site precisions than the remainder. Samples from three sites (11, 12 and 17) showed no tendency to group at any demagnetising field.

The significant site mean directions before and after cleaning are listed in Table I. A.f. cleaning has not always been successful in improving within-site precision but the end result from the treated material is that the site from the mafic border of the Complex (14) is apparently reversed with respect to the remainder, and the two sites in gabbros at Cahama (8 and 9) yield directions of magnetisation not distinguishable from the remainder and consistent with the geological inference that they are a related magmatic episode. Excluding the anomalous directions from sites 3 and 18, and reversing site 14, the significant sites combine to yield a mean direction of $D = 259°$, $I = -46°$ ($k = 7$) with a derived palaeomagnetic pole at 255°E, 3°S.

The overall precision is distinctly lower than that obtained from palaeomagnetic collections in some other large basic bodies of comparable age. The Modipe gabbro of Botswana (2700 m.y.; Evans and McElhinny, 1966) yielded a k-value of 22, the Great Dyke of Rhodesia (2,500 m.y.; McElhinny and Gough, 1963) a k-value of 33 and the Bushveld gabbro (1,950 m.y.; Gough and Van Niekerk, 1959) a k-value of 41. There are several possible reasons for this of which the two most probable are tectonic complexities and polar wander during cooling of the Complex. The structure is poorly known but by analogy with similar large Proterozoic intrusions (Bridgewater et al., 1974) may have a mushroom or funnel-shaped profile. Where developed, lineation in the anorthosite suggests that it is at least in part deformed, but in the absence of any structural indicators in the sampling region, or knowledge of whether or not deformation predated cooling through the Curie point, no tilt corrections have been applied. The presence of a close reversal of magnetisation within the body provides some evidence that this has not greatly affected the overall mean, but small tectonic tilts might contribute to reduction in overall precision.

Secondly, since the sites are distributed across a distance of 115 km it is likely that the bulk of the complex sampled would have taken more than a few million years to cool through the Curie point, or long enough for significant apparent polar wander to take place relative to the region. Also it is possible that the complex comprises more than one magmatic episode. This explanation receives some support from the distribution of pole positions examined in the last section.

4. Opaque petrography

Petrographic examination of samples from the dark troctolitic anorthosite in reflected light with oil immersion shows that the oxides fall into two groups. The first consists of anhedral grains of magnetite with diameters between 20 and 300 μ. The bulk of the grains are homogeneous but a few are subdivided by occasional

broad lamellae of ilmenite (equivalent to oxidation class 2 of Ade-Hall et al., 1968). These grains exhibit irregular outlines indicative of disequilibrium at a late magmatic stage, but nowhere are grain margins clearly preserved. They are altered peripherally and along cracks to fine granular aggregates; these are sometimes developed across ilmenite lamellae and appear to postdate their exsolution. Only hematite has been positively identified in this secondary phase.

The second group of oxides also appears to be ubiquitous in the dark troctolitic anorthosite and comprises abundant fine rods aligned along cleavage planes in the plagioclase. They have lengths up to about 40 μ and diameters down to limits of microscopic resolution ($\sim 0.5\ \mu$); they are certainly rods or needles since they are visible as pinpoints of light in cross-section and can be followed through small depths of focus with the microscope. Rods of this type have been widely reported in the literature where they have been regarded as important carriers of NRM. Evans et al. (1968) describe such rods in pyroxenes and amphiboles of the Modipe Gabbro, Botswana, and Murthy et al. (1971) and Palmer and Carmichael (1973) have described them from anorthosite bodies of the Canadian Shield. Although their small size makes identification difficult, these workes are agreed that the rods are magnetite.

5. Magnetic observations on the anorthosite

As already described, demagnetisation characteristics are similar across the sampled area of the complex and allow members of the hematite–ilmenite series to be excluded as principal remanence carriers. Temperature changes of NRM exhibited by the collection are also very uniform: all of fifteen samples examined showed uniform fall of moment with heating (Fig. 3A) and Curie points between 560 and 600°C, averaging 585°C. As is typical of slowly cooled plutonic rocks, they appear to undergo little alteration on heating, and heating curves correspond closely to cooling curves. Curie points are close to, but slightly higher than, that of pure magnetite, and demonstrate that magnetisation cannot entirely reside in a member of the magnetite–ulvöspinel series; they might be explained in terms of the small additional affect of secondary hematite.

Typical hysteresis characteristics of samples of the dark troctolitic anorthosite are shown in Fig. 3B. Most of the samples examined saturated in fields of about 2000 Oe, although values of the saturation magnetisation (J_{SAT}) vary between 500 and 3000 Oe. This is typical of the range of values of J_{SAT} observed in igneous rocks (Radhakrishnamurty et al., 1970). Roquet (1954) showed that isothermal remanence of magnetite saturates at applied fields of about 3000 Oe compared with fields of about 30,000 Oe for hematite. This result again confirms that magnetite is the main remanence carrier in the anorthosite although the technique might fail to detect small amounts of hematite. Coercivities range from 50 to 350 Oe with most values falling in the range 250–300 Oe and are high compared with most rocks in which magnetite is the principal carrier of NRM. This property (see next section) probably reflects presence of the numerous fine rods of magnetite contained within the plagioclase feldspar which lie within a size range for which coercive force increases dramatically with decrease in grain size (e.g. Nagata, 1961, p. 100); the large anhedral grains of magnetite are above this critical size range and possess lower coercivities (next section).

Lowrie and Fuller (1971) showed that saturation IRM is characteristically less stable than the weak field TRM for single domain carriers and more stable for multidomain carriers. A Lowrie-Fuller test on seven samples of the Cunene anorthosite showed that saturation IRMs were more stable to a.f. demagnetisation than NRMs (Fig. 3C) for all but two samples. This is a surprising result because most theoretical analyses explain stable TRM in terms of single-domain behaviour. It has been variously suggested that these single domains are produced by subdivision of larger grains (Strangway et al., 1968), reside in discrete grains below optical resolution (Evans and Wayman, 1970), or in optically resolvable magnetite rods of appropriate sizes (Murthy et al., 1971). The Cunene anorthosite possesses NRMs stable up to at least 1600 Oe and on general grounds it seems likely that this NRM is carried by single-domain carriers. This is further suggested by the distribution of Q-values. From data on the grain size dependence of the Koenigsberger ratio Q given by Parry (1965), Stacey (1967) proposed that a Q-value greater than one indicates a TRM held by single (or pseudosingle) domain magnetite, while Q-values greater than one indicate multidomain remanence. With the exception of samples from sides 14 and 18 the majority of samples from the Cunene anorthosite have

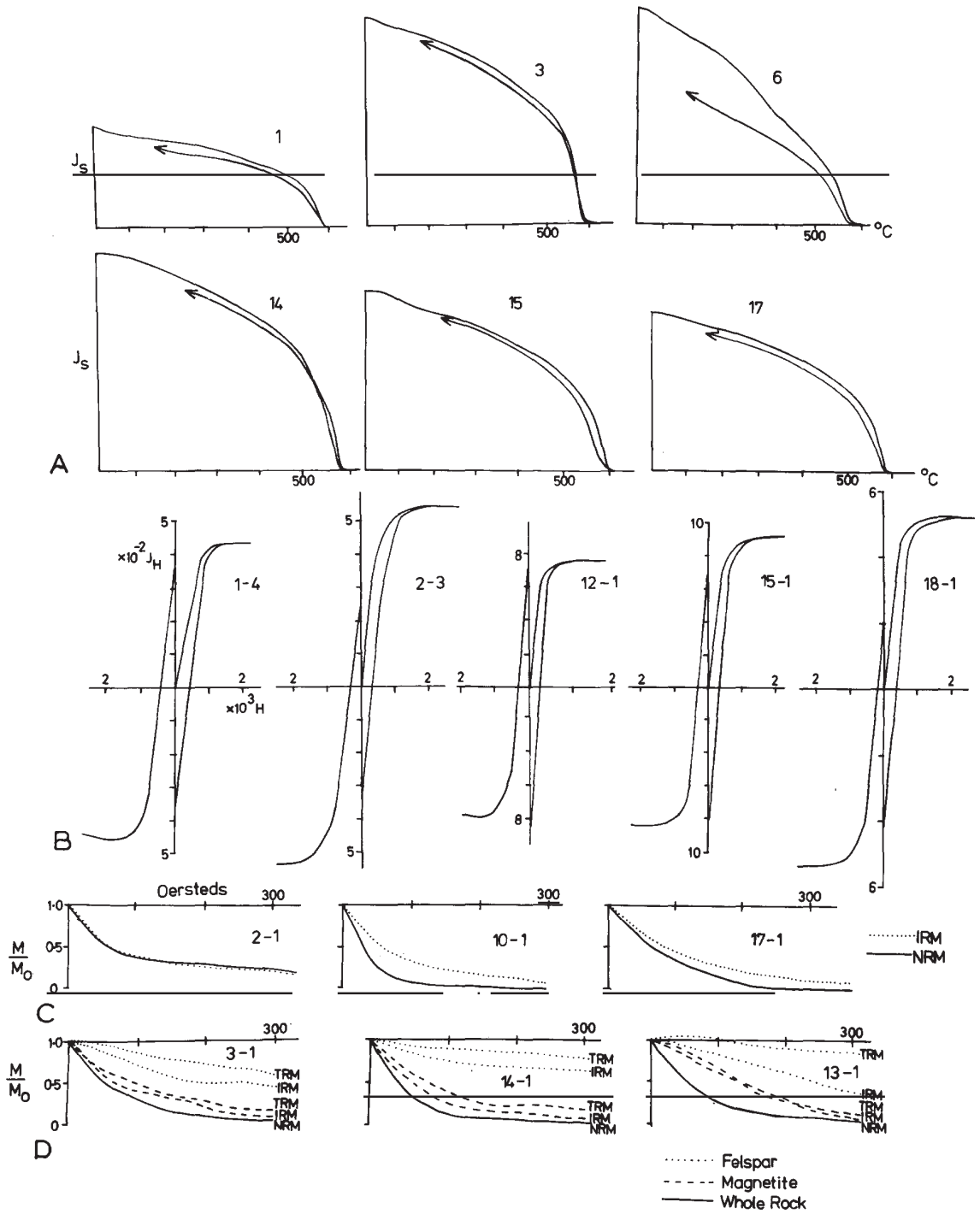


Fig. 3. A. Variation of saturation magnetisation with temperature for selected samples of the dark troctolitic anorthosite. B. Hysteresis characterics. C. Comparison of NRM and saturation IRM demagnetisation curves for whole rock samples. D. Comparison of artificial TRM and saturation IRM demagnetisation curves for separated magnetite and felspar fractions.

Q-values greater than one (Fig. 2A), although as Lowrie and Fuller have pointed out, this can only be regarded as a crude test for single-domain magnetite. There are two possible explanations for results of the Lowrie and Fuller tests: either they reflect dominance of low-field behaviour by multi-domain carriers which from the petrographic examination could be large homogeneous magnetite grains, or they could result if the total NRM is not entirely a low-field TRM and includes a component of viscous remanent magnetisation (VRM).

To examine contributions of the principal mineral phases to the magnetisation of the anorthosite, samples were crushed to 0.2 mm grain size and oxide and plagioclase feldspars separated by hand magnet and magnetic-separator. Between 3 and 9% by volume of the rocks were found to comprise magnetite grains in this way, and the two separate fractions were made up into artificial cores using dental plaster.

6. Magnetic properties of separated mineral phases

Artificial specimens were given a saturation IRM, demagnetised, and then heated to 605°C and cooled in a field of 0.5 Oe to acquire an artificial low-field TRM. In both cases the samples incorporating the magnetite fraction acquired moments of the same order but rather higher than the feldspar samples. Moments held by the feldspar samples however always possessed much higher coercivity than the magnetite samples, and after demagnetisation to 300 Oe their moments were between 3 and 10 times larger (Fig. 3D). For the first few hundred oersteds treatment, demagnetisation curves for the magnetite phase are very similar to the whole rock curves and very different from the feldspar curves. When given an artificial TRM however, both phases give single-domain Lowrie-Fuller tests and it has not proved possible to reproduce multidomain tests exhibited by the NRM. This may be due either to dominance of the total NRM by components of VRM or by alteration of the magnetite during application of the artificial TRM.

These results give a qualitative explanation for the demagnetisation curves of Fig. 2B: the large initial reductions in moment result from demagnetisation of low-coercivity components residing in magnetite grains and subsequent slight reductions correlate with high-coercivity components held by magnetite rods within the feldspar. Although both phases appear to possess single-domain carriers, the former presumably by subdivision of larger grains by ilmenite lamellae (Strangway et al., 1968) or marginal alteration and the latter as elongate rods of appropriate axial ratios (Murthy et al., 1971), the bulk of the high-coercivity stable remanence resides in magnetite in the silicate phase. This is similar to the findings of Murthy et al. (1968) from the Michikamau anorthosite and Hargraves and Young (1969) from the Lambertville diabase.

7. Interpretation of the pole positions

At the time of Simpson's paper (1970), the only control on the minimum age of the Cunene anorthosite was a single Rb/Sr age of 1290 ± 90 m.y. on a muscovite from an intrusive pegmatite (Simpson and Otto, 1967). Ages obtained from basement rocks predating the anorthosite range from 2,600 m.y. to 870 m.y. (Mendes, 1966; Cahen and Snelling, 1966). Carvaalho (1969, 1972) quotes ages of 1700 m.y. and 1300 m.y. for granite bodies intrusive into the anorthosite and a Rb/Sr age of 2079 m.y. for a biotite in granodiorite believed to be associated with the anorthosite. Dr. H.L. Allsopp (personal communication, 1973) has obtained a range of K/Ar whole-rock dates from the anorthosite between 2700 and 1100 m.y. but K/Ar and Rb/Sr ages on biotites from gabbros give ages around 1100 m.y. The age data are thus still far from conclusive but accepting the muscovite age of Simpson and Otto (1967) implies that the Complex must be older than 1290 m.y. and a well-defined group of ages ca. 1100 m.y. seems to be a consequence of thermal over-printing (H.L. Allsopp, personal communication, 1973). Subsidiary evidence for this is the situation of the Complex close to the line of the Kibaran–Orange River Belt which experienced a widespread thermal event at this time; and a thermal episode of ca. 1100 m.y. is widely recognised elsewhere in Africa (e.g. Cahen and Snelling, 1966). It is more difficult to recognise which of the older ages is close to the true age of the Complex and the scattered whole rock ages of Allsopp may be due to a combination of argon gain and argon loss although it is possible that only argon loss is involved.

The pole position derived from the Complex is currently the most reliable indicator of its true age, since both the presence of a reversal of magnetisation and the stability evidence suggest that it possesses a primary

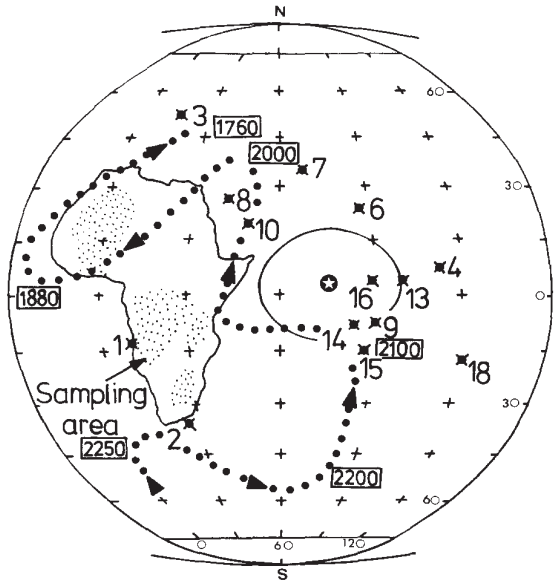


Fig. 4. Plot of pole positions derived from individual sites and from all sites. Also illustrated on this diagram is the apparent polar wander curve for Africa from 2250–1750 m.y. (after Piper et al., 1973) and the stippled area of Africa shows the cratonic regions which have remained unaffected by thermal or tectonic events since ca. 1800 m.y. (after Clifford, 1968). Circle is cone of 95% confidence for mean direction of magnetisation; note that this only includes four site virtual geomagnetic poles.

TRM acquired during cooling. Piper et al. (1973) have found that palaeomagnetic results from several parts of Africa for the interval 2250–1950 m.y. define a single apparent polar wander (a.p.w.) swathe, and indeed available Precambrian data suggest that a single a.p.w. curve applies to the whole continent from at least 2250 m.y. to the present. The antipole from the Cunene Complex (75°E, 3°N) is the only currently available Precambrian pole from the Kasai craton and falls close to the a.p.w. curve for Africa near to poles dated at ca. 2100 m.y. (Fig. 4) which is therefore regarded as the best estimate of the cooling age of the Complex. With this conclusion the reservation is made that the pole probably also lies fairly close to the African a.p.w. curve at ca. 1700 m.y. (Piper, 1974) and since the a.p.w. curve is poorly defined prior to 2300 m.y. it may conceivably correlate with this part of the curve. The pole (105°W,3°S) is in fact not widely removed from the a.p.w. curve of McElhinny et al. (1968) at ca. 2500 m.y. or close to the probable age of country rocks intruded by the Cunene Complex. However, since both of these possibilities are less likely, it would appear that the Rb/Sr age of 2079 m.y. quoted by Carvaalho is close to the cooling age of the Complex.

The low precision of the group mean direction of magnetisation may be due to apparent polar movement while the anorthosite cooled below the Curie point, and evidence for this derives from the dual distribution of most of the pole positions (Fig. 4). The group of sites falling close to the African a.p.w. curve at ca. 2100 m.y. includes sites 13, 15 and 16, and site 14 from the border zone, which all lie in the northwestern part of the anorthosite body, while four sites from the centre of the body (3, 6, 7 and 10) lie near the a.p.w. curve for Africa at ca. 2000 m.y. Sites 1 and 2 from the south-east margin of the body give pole positions not clearly related to the remainder although they may conceivably correlate with a pre-2250 m.y. section of the a.p.w. curve (Fig. 4).

The Cunene Complex is one of the suite of massive anorthosite bodies intrusion of which appears to have been confined to a limited interval of Precambrian time. Herz (1969) estimates that this interval was ca. 1750 m.y. to 1000 m.y. and crustal-temperature gradients do not appear to have been sufficiently steep after 1000 m.y. to permit their intrusion into the crust. Further age dating on the anorthosites of the U.S.S.R. in addition to conclusions of this paper would appear to extend this interval back to 2100 or 2200 m.y., but there is no evidence to suggest that their intrusion overlapped in time with injection of foliated anorthosites widely found associated with Archean granite gneiss terrains. Analysis of Precambrian palaeomagnetic data suggests that Proterozoic massive anorthosites formed along a single broad arc of crustal mobilisation (Piper, 1974) consistent with recent interpretations of tectonic data (e.g. Sutton and Watson, 1974) suggesting that the bulk of the continental crust formed a single unit in Proterozoic time subject to internal deformation along restricted zones of crustal mobility.

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